

As can be seen, the first approach is more compact (uses fewer characters), and is hence used as per convention. Similarly, Haskell also supports anonymous or lambda functions using the ‘\’ operator.

List comprehensions are a way of synthesizing a list using an expression, where each element in the list is logically related to a parameter. For example, to make a list of all numbers from 1 to 100, the following Haskell one-liner is valid:

- `[ f | f <- [1..100]]`

We can include powerful logic to modify our lists during runtime. For e.g., a list that contains the squares of all numbers present in a list `l`,

- `[f*f | f <- l]`

Similarly, a list that stores all the factors of a number `n` can be expressed as,

- `[f | f <- [1..n], n mod f == 0]`

(Here the mod operator returns the remainder after dividing `n` with `f`)

Comments in Haskell are of three types - single line comments start with `-` and end with a newline character. Multi Line comments begin with `{-` and end with `-}` and can include nested comments or code. Literate comments, a third type, aim to demarcate the code instead of the comments and start with `\begin{code}` and end with `\end{code}` as is commonly found in LaTeX. This allows Haskell programs to be easily typeset in LaTeX documents.

Monads are one of the most important features of Haskell, and can be thought of as the programmatic equivalent of an assembly line. Data that is input into a monad gets modified in stages as if it were to be on an assembly line. Formally, a monad is simply a way of describe any computation as a sequence of steps. Monads are used to define how a particular data type should behave when multiple operations on it are chained together, or how functions that are nested inside other functions should return values. Monads may be constructed for computations that handle I/O, change state of their resident variables or return multiple values.

Implementations of Haskell

The most commonly used (and full featured) Haskell compiler is the GHC (Glasgow Haskell Compiler) and is available for Windows, Mac and Linux <http://dgkit.in/fitHaskell>. Other implementations of Haskell such as lbc, Gofer and Yale Haskell are also available, but GHC is the de-facto standard for Haskell compilers. You can find the latest version of GHC on the DVD accompanying this month's FastTrack issue. Text editors such as SublimeText, Vim, IntelliJ and Eclipse support Haskell.